

"I felt Like Talking to a Wall" — A Performative Study on Early Encounters with a Social Robot

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Abstract

Early encounters between humans and social robots are brief but formative moments that shape future interaction. These moments are hard to study because they rely on pre-reflective, embodied behaviors often missed in task-driven experiments. We propose an embodied, performative methodology in which trained performers improvise with a social robot; subtle relational dynamics are then captured through micro-behavioral video analysis. In five workshops, we observed confident bids for connection, moments of micro-interruption, and adaptive strategies such as mirroring and attunement that repaired disrupted interactions. This lens reveals how early relationality is enacted and maintained, offering HCI researchers new methodological and design insights for making social robots more legible and responsive during first contact.

CCS Concepts

- **Human-centered computing** → Empirical studies in HCI;
- **Social and professional topics** → *User characteristics*;
- **Computer systems organization** → Robotic autonomy.

Keywords

Embodied Interaction, Relational Design, Human-Robot Interaction, Artistic-inspired Research, Intimate Computing

ACM Reference Format:

Michaela Honauer and M. Birna van Riemsdijk. 2026. "I felt Like Talking to a Wall" — A Performative Study on Early Encounters with a Social Robot. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3772363.3798667>

1 Introduction

Early encounters between humans and social robots are brief and delicate moments that can influence how relationships unfold later on. Prior work has richly examined longer engagement while first moments of contact remain underexplored. However, human–human research shows early impressions can shape subsequent interaction quality [8]. Early encounters are often pre-reflective what can be explained with the *dual process model* that differentiate between the intuitive (pre-reflective) system and the cognitive system [9].

In dance literature, such pre-reflection is referred to as *body knowledge* what means that people lean on bodily orientation through gaze and timing, not on deliberate reasoning [3, 17]. Capturing such unconscious dynamics requires methods that reveal relevant micro-behaviors.

In this paper, we introduce an embodied and performative methodology for studying how relationality is enacted in the very beginning of human–robot contact. We draw on perspectives that see sociality as co-constructed in situ rather than a fixed characteristic of a robot or user [5, 10]. Practically, we invite trained performers to improvise with a social robot, using movement to probe attention, proximity, timing, and mutual attunement. We evaluate these sessions through micro-behavioral video analysis to make subtle, embodied processes visible [6, 16].

Our goal is to show how artistic techniques combined with close interaction analysis provide a methodological lens for studying relational dynamics in early human–robot encounters. Our methodological approach complements conventional HRI study designs and helps to reveal phenomena that are based on *body knowledge* [3, 17] and that might otherwise remain hidden.

2 Background

2.1 Early encounters as embodied, pre-reflective interaction

Early contact between humans and robots unfolds through pre-reflective, bodily modes of knowing—movement, gaze, posture, and orientation—rather than deliberate reasoning [3, 17]. In human–human contexts, these first micro-moments can shape relational trajectories [8, 19]. Transferring this insight to HRI suggests that understanding early human–robot relationality requires methods that capture micro-behaviors as they emerge. Instead of treating sociality as a fixed property of either human or robot, recent work conceptualizes it as co-constructed in situ [5, 10], underscoring the need for approaches that observe how relationships are enacted moment-to-moment.

2.2 Performative methods as a lens on relational enactment

Recent research has called for more selective integration of performing arts techniques into HCI research [20]. In dance studies, movement is understood as a form of knowledge—a way to explore relationships, sense others, and negotiate shared space [3, 17]. These traditions emphasize that meaning in interaction arises through mutual attunement, timing, and embodied presence. Applying this perspective to HRI allows participants to probe a robot's relational



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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798667>

capacities through improvisation rather than fixed tasks. This aligns with calls in HCI and HRI to explore alternative, non-task-based methods that move beyond assumptions about social cues aligning with human norms and to open space for robot-specific relational interactions [7, 12]. Furthermore, it is characteristic of performance arts techniques that they do not necessarily have to be problem- and solution-based, but rather enable novel explorations [4].

2.3 Micro-behavioral analysis for capturing fine-grained dynamics

To study these processes in detail, micro-behavioral video analysis offers tools to track proxemics, gaze, orientation, and subtle timing shifts [6, 16]. Such fine-grained observation makes visible small interruptions and reconnections—brief gaze breaks, orientation shifts, or re-engagement attempts—that shape the flow of early interaction. Combined with performative exploration, this analytic approach creates a methodological framework well-suited to uncover how relationality is enacted, disrupted, and recalibrated in the first moments of human–robot contact.

3 Methodological Approach

3.1 Aim and rationale of the study design

Our aim was to examine how relationality is enacted during the initial moments of contact between humans and a social robot. Therefore, we chose an embodied, performative approach to investigate early encounters. This methodological approach was combined with micro-behavioral video analysis. A studio-based, performative setup allowed us to bring tiny interaction details (gaze, posture, proximity, timing) to the foreground and to repeatedly probe them through improvisations with a social robot. The studio setting also minimized uncontrolled disruptions typical of field contexts. Trained performers served as movement experts capable of intentionally exploring and modulating relational tactics (e.g., mirroring, circling, lowering to eye-level) to make relational dynamics legible during interaction.

3.2 Setting and procedure of the data collection

Workshops were held in a dance/theater rehearsal room to ensure a clear line of sight for cameras and unobstructed movement. Each session lasted around 3 hours and followed the same structure:

- (1) Briefing and consent (including short demographics).
- (2) Performative warm-ups to establish embodied readiness.
- (3) Improvisational movement experiments, including brief breaks and immediate flash-light interviews.
- (4) Closing group conversation on impressions, strategies, perceived limitations, and relational dynamics.

Improvisations were paced by the length of one song (around 3–5 minutes). Performers were asked to continuously interact with the robot, to seek its attention, and to proceed at their own pace and comfort, using free improvisation and, when desired, pre-rehearsed choreography. Exercises typically began with one performer interacting with the robot; later, a second or third might join. For the analysis highlighted in this paper, we focus exclusively on the initial one-to-one moments (a performer's first encounter with the robot). Figure 1 provides an impression of the room layout during

an example exercise; the control performer operating the robot Pepper was always visible to the group.



Figure 1: Left: Study setup in a rehearsal room: Robot in the center, performers dancing around trying to grasp its attention. Another performer positioned visibly on the side, controlling the robot's movements through skeleton tracking with a Kinect sensor. Right: Robot Pepper; we did not use its voice functions and the touch display on its chest.

3.3 Performing with the robot

We used the social robot Pepper (Figure 1, right). Its voice and touch-display were not used to foreground embodied coordination rather than verbal exchange. One performer—visibly positioned—controlled Pepper's upper body, head, arms, and hands via camera-based skeleton tracking (Microsoft Kinect 2.0) as shown in Figure 1 (left). Their movements were directly mapped to Pepper's corresponding joints. Pepper's Autonomous Life feature [18] remained activated to introduce a small degree of spontaneity (e.g., looking around when not focused on a single person; reacting to environmental sounds). This combination created an encounter that was responsive yet not fully scripted, supporting situational attunement and timing-sensitive bids.

3.4 Participants

Across five workshops, 26 performers participated from breakdance/ hip-hop, modern dance, and improv theater. Participants were mainly students (average age ≈ 22 years), with self-reported expertise from beginner to advanced and mixed prior robot exposure. Table 1 (in the Appendix) gives an overview of the participants. Recruitment occurred via the university's arts and culture associations and sessions took place during regular rehearsal times. Participation was voluntary and informed. We provided refreshments and a €25 voucher per participant.

3.5 Data collection

All improvisational exercises were video-recorded to document fine-grained interaction details such as gaze, posture, orientation, proximity, and timing. Two camera perspectives ensured that both the performer and the robot remained visible throughout the improvisation so that the footage could later be examined at a suitable level of granularity for micro-behavioral analysis. After each improvisational exercise, we conducted short flash-light interviews to capture performers' immediate impressions of interacting with the robot. At the end of each workshop, a group conversation was also recorded, providing additional context about perceived relational dynamics and strategies. All interviews were audio-recorded.

For the analysis presented in this paper, we drew on ten video recordings of solo interactions between a single performer and Pepper, ten corresponding flash-light interviews, and five group discussions—one from each workshop. These combined materials provided both a detailed behavioral record and verbal accounts of performers' experiences, enabling us to compare what occurred in the interaction with how performers perceived and interpreted these early encounters from an individual perspective.

3.6 Analysis strategy

We conducted a micro-behavioral analysis of the solo-interaction videos, combining elements of the *Multimodal Interaction Analysis Framework* [11] (to trace proxemics, posture, gaze) with a *4-Level Interaction Concept Analysis* [16] (to situate micro-moves within context and perceived impact). Three researchers (two were not co-authors and specifically instructed) transcribed and coded the videos to extract interaction changes and to group them into themes (e.g., seeking attention; proximity and touch; adapting to the other; moments of disconnection). Audio from the interviews was transcribed verbatim and analyzed using *Open Coding* [14] to surface perceived limitations, unmet expectations, and adaptive strategies. We then compared the video recordings (what happened in the interaction) with the performers' accounts (how it felt), which allowed us to articulate distinctive patterns of initiation, disruption, and recalibration during first contact.

4 Findings

Our analysis of the first encounters between the performers and Pepper reveals a coherent relational trajectory: (1) **confident initiation**, (2) **micro-disconnections**, and (3) **embodied recalibration**. This arc unfolds through pre-reflective bodily engagement and illustrates how relational meaning is explored, disrupted, and rebuilt in short time-frames.

4.1 Confident initiation: immediate bids for connection

Across all workshops, performers approached Pepper with direct and attuned movement. They moved straight toward the robot, maintaining immediate focus to establish interaction through sustained gaze and forward-oriented posture (Figure 2, right). Sonic cues emerged spontaneously: improv actors shouted or bended down talking to the robot, while dancers stamped on the ground or clapped hands to draw attention (Figure 2, left). Performers also reduced distance rapidly; all of them came very close, often within a range of 0.5–1 m, and most of them approached Pepper from behind (Figure 2, bottom). These behaviors indicate intuitive, embodied attempts to initiate connection with a non-human social partner.

4.2 Micro-disconnections: interruptions in emerging interactions

Performers quickly encountered breaks in reciprocity when Pepper failed to react as expected. The robot frequently turned away, appeared distracted, or presented what one performer called a "dead stare with a smile". Such moments produced subtle irritation, visible through the micro-analysis as looking away for milliseconds



Figure 2: Left: Performers stamped with their feet, clapped hands, or talked to the robot. Right: Performers were attuned to the robot, bending towards it or holding arms open. Bottom: Performers came closer to the robot than culturally normed for humans, often approaching it safely from behind.

or small posture adjustments (Figure 3, top). Performers explicitly articulated their felt experiences in the interviews: "You expect a reaction and... there isn't any", "I felt like I'm talking to a wall", or "the robot takes like 5 seconds to react... [but] the moment has long passed away". These micro-disconnections highlight the fragile timing required for early attunement.

4.3 Embodied recalibration: adapting tactics to sustain engagement

Rather than withdrawing, performers adjusted their relational tactics. Many used mirroring and imitation to re-establish connection—e.g., through copying what Pepper is doing or imitating its mechanical movements (Figure 3, left). Others adapted spatially by lowering themselves to Pepper's height—e.g., through kneeling or in a squat—to get on eye-level and align with its gaze and movement range (Figure 3, right). Performers described this adaptation as necessary: "that you adjusted to the robot instead of the robot adjusting to you" and "I tried to connect it to make it equal as me, but... I first



Figure 3: Top: The micro-analysis reveals that performers sometimes look away for milliseconds. Left: The performers were imitating the robot’s movements, mainly with their arms and upper body. Right: The performers were often lowering their bodies and taking on postures that were literally on Pepper’s eye-level.

had to unlearn... then learn what you can do with the robot”. The interaction remained enjoyable yet asymmetric, and was described as “one-sided”.

5 Discussion and implications

Our performative, embodied method combined with micro-behavioral analysis proved useful for uncovering pre-reflective relational processes that are difficult to access through conventional task-driven studies. The performers’ immediate bodily attunement aligns with phenomenological accounts of *embodied knowing* [3, 17] and confirms that early encounters rely on intuitive, sensorimotor coordination rather than explicit reasoning. The method also supports relational HRI perspectives that view sociality as co-constructed in situ [5, 10], which became evident in how performers shaped the interaction through gaze, proximity, and timing, even when the robot did not respond as expected.

At the same time, our findings nuance existing assumptions. While prior work suggests humans often maintain human-normed distances [1] with robots [15], performers in our study consistently reduced proximity and initiated touch, echoing results from virtual-agent research where people approach non-human agents more closely than humanoid agents [2]. Similarly, the shift from initial humanized expectations to later recalibration resonates with a study that found anthropomorphism is a dynamic process rather than a static state [13]. Our analysis shows how this shift unfolds

through tiny interruptions in attention and bodily adaptation, expanding previous accounts by offering details of how such recalibration is enacted.

Finally, performers’ frustration with delayed or absent reactions highlights the importance of timing-sensitive cues for maintaining early mutuality. Across workshops, participants described instances where the robot looked away too long, reacted too slowly, or failed to acknowledge their relational bids altogether. Our observations suggest that even minimal, well-timed micro-responses—such as a brief re-orientation of the head or torso toward the performer, a short postural shift signaling acknowledgment, or a momentary modulation of movement rhythm—could sustain the reciprocity loop more effectively than static anthropomorphic signals. Such timing-based *micro-acknowledgments* align with what performers naturally sought and adapted to in the interaction, and point toward early-stage design opportunities for social robots to support smoother, more legible initiation of mutual engagement.

6 Conclusion

This study introduced an embodied, performative methodology for examining how relationality is enacted in the first moments of human–robot contact. By inviting performers to explore the robot through improvisation and analyzing the resulting interactions with micro-behavioral techniques, we were able to access pre-reflective, moment-to-moment dynamics that typically remain hidden in task-driven HRI studies. The findings reveal a consistent arc of confident initiation, micro-disconnection, and embodied recalibration, demonstrating how humans actively work to build, repair, and sustain early mutuality with a social robot.

These insights show that early relationality depends less on anthropomorphic appearance and more on timing, responsiveness, and embodied legibility. They further suggest that designing for smooth early interaction may involve creating robot-specific, well-timed micro-cues rather than replicating signals that build on human norms. Taken together, our methodological and empirical contribution points toward a richer understanding of how social robots can invite, acknowledge, and sustain human bids for connection from the very beginning of an encounter.

Acknowledgments

Many thanks to the participating performer. We are also grateful for the support of our colleagues from the Human Media Interaction Group/ University of Twente: Hideki Garcia Goo with whom we conducted this study; and Luc Schoot Uiterkamp, Camille Sallaberry and Edwin Dertien who helped with the technical preparations. Thankfully, we were also able to discuss this work with ethicist, philosopher of technology, and researcher Anna Puzio.

This research has been approved by the ethical committee of the faculty Electrical Engineering, Mathematics and Computer Science at University of Twente with number 230360.

Generative AI (Microsoft 365 Copilot) was used as a writing aid by the first co-author to help phrase sentences from author-provided notes, to revise and shorten earlier drafts. All research ideas, the literature review, and the structure were developed by the authors independently. The authors take full responsibility for all content, claims, and interpretations in this paper.

References

- [1] Nancy L. Ashton, Marvin E. Shaw, and Annette Pearce Worsham. 1980. Affective reactions to interpersonal distances by friends and strangers. *Bull. Psychon. Soc.* 15, 5 (May 1980), 306–308. doi:10.3758/BF03334541
- [2] Jeremy N. Bailenson, Jim Blascovich, Andrew C. Beall, and Jack M. Loomis. 2001. Equilibrium Theory Revisited: Mutual Gaze and Personal Space in Virtual Environments. *Presence: Teleoperators & Virtual Environments* 10, 6 (Dec. 2001), 583–598. doi:10.1162/105474601753272844
- [3] Betty Block and Judith Lee Kissell. 2001. The Dance: Essence of Embodiment. *Theoretical Medicine* 22 (2001), 5–15. <https://nyuskirball.org/wp-content/uploads/2018/02/The-Dance-Essence-of-Embodiment.pdf>
- [4] Sarah Fdili Alaoui. 2019. Making an Interactive Dance Piece: Tensions in Integrating Technology in Art. In *Proceedings of the 2019 on Designing Interactive Systems Conference (DIS '19)*. Association for Computing Machinery, New York, NY, USA, 1195–1208. doi:10.1145/3322276.3322289
- [5] Petra Gemeinboeck and Rob Saunders. 2022. Moving beyond the mirror: relational and performative meaning making in human–robot communication. *AI & Soc* 37, 2 (June 2022), 549–563. doi:10.1007/s00146-021-01212-1
- [6] Roger Haeussling. 2009. Video analysis with a four-level interaction concept: A network-based concept of human-robot interaction. In *Video interaction analysis: methods and methodology*, Ulrike Kissmann (Ed.). Peter Lang, Frankfurt am Main ; New York, 107–131. OCLC: ocn311755462.
- [7] Marc Hassenzahl, Jan Borchers, Susanne Boll, Astrid Rosenthal-von der Pütten, and Volker Wulf. 2021. Otherware: how to best interact with autonomous systems. *interactions* 28, 1 (2021), 54–57. doi:10.1145/3436942
- [8] Lauren J. Human, Gillian M. Sandstrom, Jeremy C. Biesanz, and Elizabeth W. Dunn. 2013. Accurate First Impressions Leave a Lasting Impression: The Long-Term Effects of Distinctive Self-Other Agreement on Relationship Development. *Social Psychological and Personality Science* 4, 4 (July 2013), 395–402. doi:10.1177/1948550612463735
- [9] Daniel Kahneman. 2003. A perspective on judgment and choice: Mapping bounded rationality. *American Psychologist* 58, 9 (2003), 697–720. doi:10.1037/0003-066X.58.9.697
- [10] Waki Kamino, Malte F. Jung, and Selma Sabanović. 2024. Constructing a Social Life with Robots: Shifting Away From Design Patterns Towards Interaction Ritual Chains. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*. Association for Computing Machinery, New York, NY, USA, 343–351. doi:10.1145/3610977.3634994
- [11] Ulrike Kissmann (Ed.). 2009. *Video interaction analysis: methods and methodology*. Peter Lang, Frankfurt am Main ; New York. OCLC: ocn311755462.
- [12] Matthias Laschke, Robin Neuhaus, Judith Dörrenbächer, Marc Hassenzahl, Volker Wulf, Astrid Rosenthal-von der Pütten, Jan Borchers, and Susanne Boll. 2020. Otherware needs Otherness: Understanding and Designing Artificial Counterparts. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society (NordiCHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–4. doi:10.1145/3419249.3420079
- [13] Séverin Lemaignan, Julia Fink, and Pierre Dillenbourg. 2014. The dynamics of anthropomorphism in robotics. In *Proceedings of the 2014 ACM/IEEE international conference on Human-robot interaction (HRI '14)*. Association for Computing Machinery, New York, NY, USA, 226–227. doi:10.1145/2559636.2559814
- [14] Saul McLeod. 2024. Open Coding In Qualitative Research. doi:10.13140/RG.2.2.32424.89608 Publisher: Unpublished.
- [15] Jonathan Mumm and Bilge Mutlu. 2011. Human-robot proxemics: physical and psychological distancing in human-robot interaction. In *Proceedings of the 6th international conference on Human-robot interaction (HRI '11)*. ACM, New York, NY, USA, 331–338. doi:10.1145/1957656.1957786
- [16] Sigrid Norris. 2004. *Analyzing Multimodal Interaction: A Methodological Framework*. Routledge, Taylor & Francis Group, New York and London.
- [17] Jaana Parviainen. 2002. Bodily Knowledge: Epistemological Reflections on Dance. *Dance Res. J.* 34, 1 (2002), 11–26. doi:10.2307/1478130
- [18] SoftBank. n.d. Understanding Autonomous Life settings — Aldebaran 2.5.11.14a documentation. http://doc.aldebaran.com/2-5/family/pepper_user_guide/life_pep.html
- [19] Sophie Wohltjen and Thalia Wheatley. 2021. Eye contact marks the rise and fall of shared attention in conversation. *Proc. Natl. Acad. Sci. U.S.A.* 118, 37 (Sept. 2021), e2106645118. doi:10.1073/pnas.2106645118
- [20] Qiushi Zhou, Cheng Cheng Chua, Jarrod Knibbe, Jorge Goncalves, and Eduardo Velloso. 2021. Dance and choreography in HCI: A two-decade retrospective. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3411764.3445804

A Appendix

Table 1: Overview of the participating performers and their experience levels. All information is based on self-reported data provided by the performers.

Workshop-Participant No.	Performing Arts Style	Expertise Level	Age	Gender	Nationality	Experience with Robots
1-01	Breakdance/ Hip-Hop	Intermediate	25	Male	Dutch	No
1-02	Breakdance/ Hip-Hop	Beginner	20	Male	Czech	Yes
1-03	Breakdance/ Hip-Hop	Intermediate	21	Female	Dutch	No
1-04	Breakdance/ Hip-Hop	Intermediate	27	Male	Dutch	Yes
1-05	Breakdance/ Hip-Hop	Beginner/Intermed.	22	Male	Dutch	No
1-06	Breakdance/ Hip-Hop		23	Male	Dutch	No
2-01	Improv Theatre	Intermediate	25	Female	Dutch	No
2-02	Improv Theatre	Intermediate	27	Male	Dutch	Yes
2-03	Improv Theatre	Beginner	30	Female	German	Yes
2-04	Improv Theatre	Advanced	26	Male	Dutch	Yes
2-05	Improv Theatre	Intermediate	23	Female	German	No
2-06	Improv Theatre	Intermediate	19	Male	Dutch	Yes
2-07	Improv Theatre	Beginner	33	Female	Greek	Yes
3-01	Modern Dance	Intermediate	24	Female	German	No
3-02	Modern Dance	Advanced	18	Female	Dutch	Yes/No
3-03	Modern Dance	Intermediate	21	Female	Russian/Latvian	No
3-04	Modern Dance	Advanced	20	Female	Estonian	No
3-05	Modern Dance	Advanced	19	Female	Dutch	Yes
3-06	Modern Dance	Advanced	23	Female	Latvian	No
3-07	Modern Dance	Beginner	21	Female	Japanese	No
4-01	Breakdance/ Hip-Hop	Advanced	23	Female	Italian	No
4-02	Breakdance/ Hip-Hop	Beginner	21	Male	Dutch	No
4-03	Breakdance/ Hip-Hop	Intermediate	25	—	Dutch	Yes
4-04	Breakdance/ Hip-Hop	Intermediate	20	Male	Czech	Yes
4-05	Breakdance/ Hip-Hop	Intermediate	39	Male	Mexican	No
5-01	Modern Dance	Advanced	22	Female	Latvian	No